

IAN SNIDER

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Robotist designing autonomous aerial vehicles for the field.

EDUCATION

Ph.D. Nuclear Engineering - Robotics - University of California, Berkeley		Aug 2025 - present
B.S. Mechanical Engineering - Washington University in St. Louis	4.00/4.00	May 2025
B.A. Physics, Math minor - Truman State University	3.89/4.00	Dec 2024

EXPERIENCE

Lawrence Berkeley National Laboratory - Lab Affiliate, Berkeley, CA Aug 2025 - present

- Developed vision-based control for TARES, a hexacopter system steering a 10 m tether-suspended radiation detector
- Designed and implemented safety-critical software infrastructure for autonomous systems
- Architected a communication pipeline for GCS, transmitter, companion computer, and Cube Blue flight controller
- Executed autonomous flight tests at LBNL and Richmond Field Station under FAA Part 107
- Field-tested TARES at a radiological site in a demonstration for senior stakeholders

PUBLICATIONS

I. Snider, B. J. Quiter, E. Rofors, M. W. Mueller. "Vision-Based Control of a Tether-Suspended Aerial Radiation Sensing Payload." *arXiv preprint [arXiv:2609.27219](https://arxiv.org/abs/2609.27219)*, 2026. Submitted to IEEE ICRA 2027.

PROJECTS

Micro Mocap Low-Cost Mocap System for Small Drones - Developer Summer 2026

- Built a motion capture state estimation system from ~\$10 PS3 Eye cameras tracking IR LEDs
- Used it to fly an ~\$80 custom racing drone with stable maneuvering

Sniff Consensus Protocol for Noisy Multi-Agent Formations - Developer Fall 2025

- Designed a consensus protocol for multi-agent formations on random graphs with noisy communication
- Simulated Erdős-Rényi agent graphs and analyzed stability using ROS2 and Python

WashU Robotics MATE ROV & MARINER Projects - Mechanical Lead 2023 - 2025

- Designed and tested an autonomous underwater vehicle (AUV)
- Prototyped 1D hovering dive control using IMU and depth state estimation with a Kalman filter
- Built a chassis and buoyancy engine

Crazyflie 2.1 cascaded PID/PD Control System - ESE 4481 Course Project Spring 2025

- Designed a control system for a Crazyflie 2.1 quadcopter and achieved stable autonomous flight
- Simulated response and stability margins in Simulink and MATLAB; programmed the embedded controller in C

ADDITIONAL RESEARCH

Lawrence Livermore National Laboratory - Graduate Intern, Livermore, CA Jun 2025 - Dec 2025

- Developed CRISP, an HPC pipeline that automated compilation and benchmarking of ~3420 reactor simulations

Brookhaven National Laboratory - Student Researcher, Upton, NY Summers 2022 - 2024

- Developed BRR (Bayesian Resonance Reclassifier), a machine-learning classifier for nuclear data on HPC clusters

Truman State University - Student Astronomy Researcher, Kirksville, MO 2021 - 2022

- Calculated trajectories for ~3000 Starlink satellites to optimize telescope viewing plans and developed a GUI

SKILLS

- **Robotics:** ArduPilot, MAVLink, ROS2, OpenCV, PlatformIO
- **Programming:** C, C++, Python, Bash, Lua, MATLAB, LaTeX, Typst, Vimscript
- **Tools:** Simulink, SolidWorks, Slurm, Git, Linux